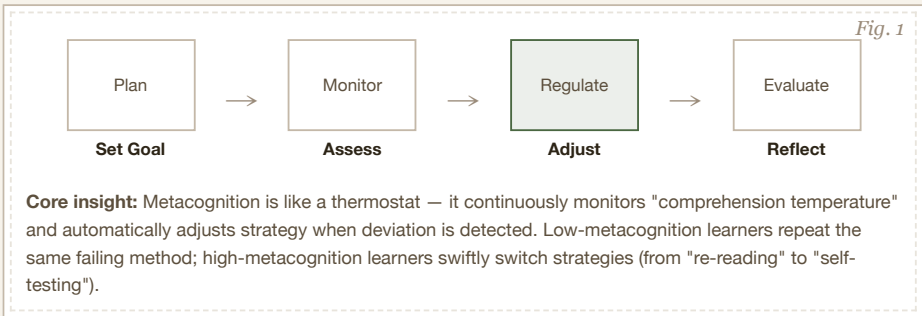


Metacognition: The Brain's Dashboard

元认知：大脑的仪表盘

Thinking about thinking — knowing what you know, what you don't, and how to improve · Printable Plate

I. THE THERMOSTAT · THE METACOGNITIVE LOOP



II. OBSERVATIONS · OBSERVATIONS

Fig. 2

Obs. 1 The Dunning-Kruger Effect — The least competent are often the most confident (Dunning et al., 2003). Metacognitive ability itself requires knowledge: the less you know, the less you know you don't know. This explains why beginners consistently underestimate difficulty.

Obs. 2 Illusion of Explanatory Depth — Most believe they understand how bicycles, zippers, or locks work until asked to explain step-by-step (Rozenblit & Keil, 2002). "Familiarity" ≠ "understanding" — this is where metacognition fails.

Obs. 3 Calibration predicts learning — Students who accurately predict exam performance score higher (Dunlosky & Rawson, 2012). Those with >20% calibration bias perform significantly worse even with equal study time. Metacognitive accuracy matters more than time spent.

Obs. 4 Metacognitive strategies separate experts from novices — Experts don't just have more knowledge; they know better when they do and don't understand. When blocked, experts step back, restructure, and seek alternative paths; novices persist with the same failing method.

Related Books

Why Don't Students Like School? Daniel Willingham | Make It Stick Brown, Roediger & McDaniel

III. METHODS · PRACTICAL METHODS

Fig. 3

- 1 Predict before studying** — Before opening a chapter, ask: "Do I think this has 3 or 8 core concepts?" Compare after reading. This simple "predict-verify" loop significantly improves metacognitive calibration (Dunlosky et al., 2013).
- 2 Self-explanation (Elaborative Interrogation)** — After each paragraph, ask: "Why is this true?" "What does this connect to?" "What if conditions changed?" Self-explanation surfaces hidden gaps in understanding.
- 3 Error analysis, not error correction** — After mistakes, don't just check the right answer. Analyze: "Why did I err?" "Where did my reasoning break?" "Was this conceptual or careless?" Turn errors into metacognitive training data.
- 4 Think aloud** — Verbalize your thoughts while solving problems. This forces vague intuitions into explicit language; when words get stuck, understanding has gaps. One of cognitive science's most effective metacognitive training methods.
- 5 Weekly learning journal** — Spend 10 minutes weekly answering: "What did I learn?" "What worked/didn't?" "What will I adjust next week?" This simple retrospective significantly improves metacognitive ability within months.

** Metacognition is cognitive science's most underrated "superpower" — unlike working memory, it isn't constrained by innate limits but can be continuously improved through training. Effects remain significant after 6 months (Hattie, 2009).*

ople fail to recognize their own incompetence. *Current Directions in Psychological Science*, 12(3), 83-87. | Rozenblit & Keil (2002).
uces underachievement. *Learning and Instruction*, 22(4), 271-280.

